Fakultas Informatika
School of Computing
Telkom University

Aplikasi Berbasis Platform

CSS & Bootstrap

CSS

- Stands for Cascading Style Sheet

- Main purpose of CSS is to separate the style (presentation) with the content (information)

Advantages of using CSS

- Rich styling
- One style for many pages
- Easier website design and maintenance

I will always find you
I WILL ALWAYS FIND YOU
FONTS MATTER.

How to use CSS in HTML

- Inline
  - Directly in HTML tag as an attribute
  - `<p style="color:green;border:solid"></p>`

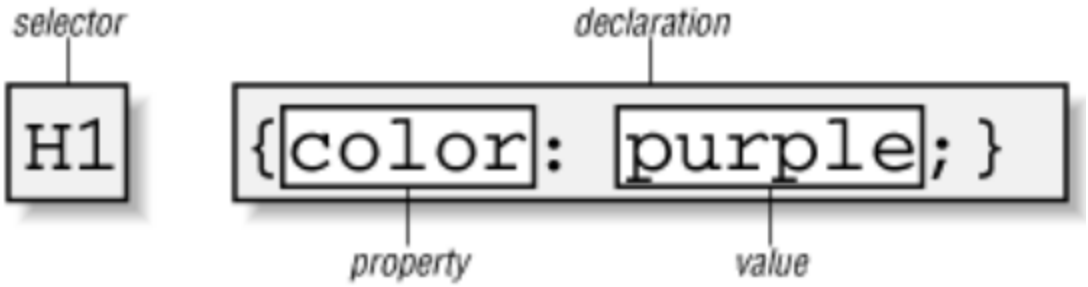
- Embed
  - Inside HTML document as a tag
  -
```
<style type="text/css">
    p {
        color:green;
        border:solid
    }
</style>
```

How to use CSS in HTML

- Linked / External file
  - Inside HTML document header (<head> tag)
    - Tag <link>
      - `<link rel="stylesheet" type="text/css" href="css1.css" />`
    - or @import
      -
```
<style type="text/css">
    @import url('css1.css');
</style>
```

CSS scripting rules

- Inline
  - <[tag] style="[property]:[value]"></[tag]>
- Embed & Linked



  - Selector: HTML element, class, or id
  - Property: styling attribute
  - Value: the value specified

CSS scripting rules

- Spacing, Tab, Return
  - just like HTML

- Comments
  - Like C language. Code comment is inserted between /* .. */

- Grouping
  - Selector : `h1, h2, h3 {font-family: arial}`
  - Declaration : `h1 {font-size: 14pt; color: blue}`
  - Value : `h1 {font: bold 12pt arial}`

Selector

- HTML element (tag)

```
h1 {color: green}
p  {font-size: 18pt}
```

- Class
  - HTML: `<p class="mhs">teks</p>`
  - CSS: `.mhs {color: gray; font: Arial}`

- ID
  - HTML: `<p id="mhs1">teks</p>`
  - CSS: `#mhs1 {color: gray; font: Arial}`

Selector

- Rule is contextual
  - Rule will be applied only if certain conditions are met. Ex:

    ```
    h2 i {color: navy}
    ```

    => only applies for all <i> tags that are inside <h2> tag
  - If HTML is:

    ```
    <h2>judul 1 <i>judul 2</i> </h2>
    ```

    which text has the navy color?

Inheritance

- If styling isn't defined, then the styling defined in its parent element will be used

- Example:

```
body {color: purple}
h2 {font-weight: bold}
```

what color will appear for <h2>?

Specificity

- To determine which styling is used if more than one styling is defined for the same element (selector)
- Highest specificity value will be used
- Order of specificity:
  - Inline style
  - ID (value: 100)
  - Class (Value: 10)
  - HTML tag/element (value: 1)

Specificity

- Example 1:
  - CSS

```
.par {color: blue}
p {color: green}
#p1 {color: orange}
```

  - HTML

```
<p class="par" id="p1" style="color:red">Hi</p>
```

what color will appear for text ‘Hi’?

Specificity

- Example 2:
  - CSS

```
.par {color: blue}
p {color: green}
#p1 {color: orange}
```

  - HTML

```
<p class="par" id="p1">Hi</p>
```

what color will appear for text ‘Hi’?

Specificity

- Example 3:
  - CSS

```
.par {color: blue}
p {color: green}
```

  - HTML

```
<p class="par" id="p1">Hi</p>
```

what color will appear for text ‘Hi’?

Pseudo-class

- Extended tag that can be used as selector
- Usually to make styling if specific action is taken (event)
- Example: <a>

```
a:link      { color: red }
a:visited   { color: green }
a:hover     { color: blue }
```

Pseudo-element

- Extended tag can be used as selector

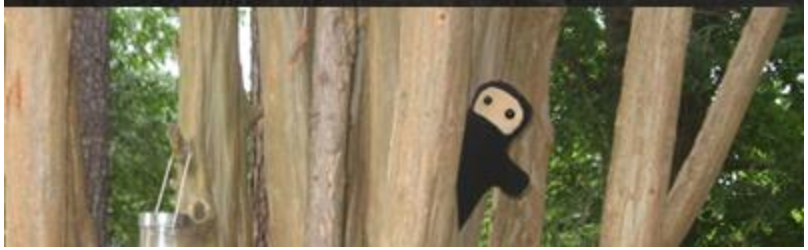
- Usually to make styling of specific text condition

- Example: <p>

```
p:first-letter {font-size: 200%}
p:first-line {color: green}
```

THE FIRST few words of an article in The Economist.

```
#ninja {
    visibility:hidden;
}
```

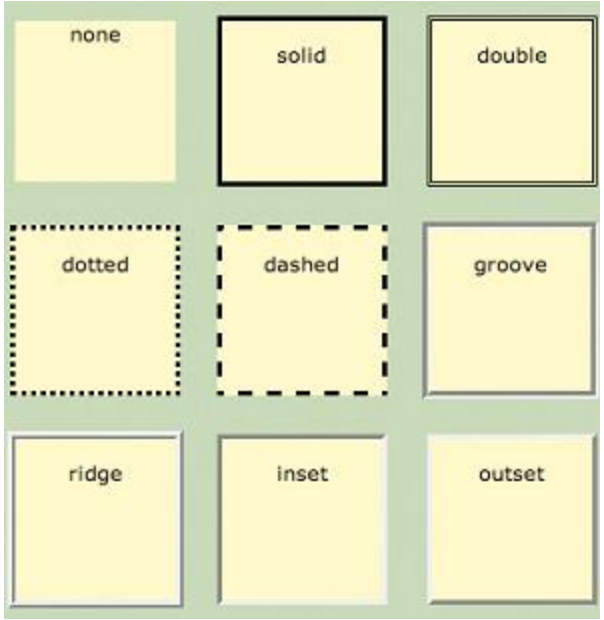


```
#titanic {
    float:none;
}
```

CSS Box Model

- Content - The content of the box, where text and images appear
- Padding - Clears an area around the content. The padding is transparent
- Border - A border that goes around the padding and content
- Margin - Clears an area outside the border. The margin is transparent

CSS for Box



A
```
-webkit-box-shadow: 6px 6px #666;
-moz-box-shadow: 6px 6px #666;
box-shadow: 6px 6px #666;
```

B
```
-webkit-box-shadow: 6px 6px 5px #666;
-moz-box-shadow: 6px 6px 5px #666;
box-shadow: 6px 6px 5px #666;/* 5 pixel blur */
```

C
```
-webkit-box-shadow: 6px 6px 5px 10px #666;
-moz-box-shadow: 6px 6px 5px 10px #666;
box-shadow: 6px 6px 5px 10px #666;/* 5px blur, 10px spread */
```

p { width: 200px; height: 100px; background: darkorange; }

border-top-right-radius: 50 px;

border-top-left-radius: 1 em;
border-top-right-radius: 2 em;
border-bottom-right-radius: 1 em;
border-bottom-left-radius: 2 em;
~or~
border-radius: 1 em 2em;}

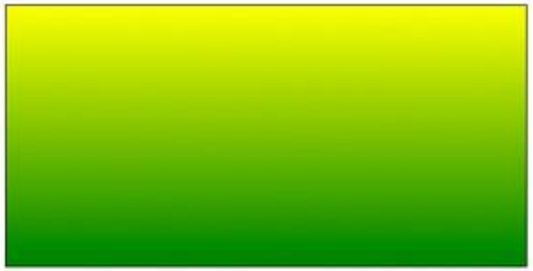
border-radius: 5px 20px; 40px 60px;

border-radius: 1em;

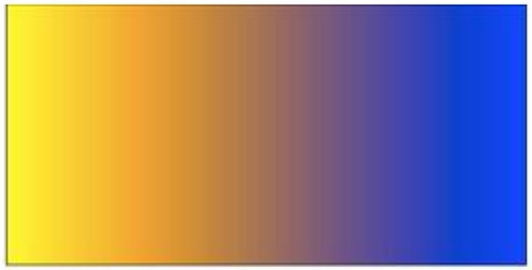
border-radius: 50 px;

Advanced CSS: Gradient BG Color

```
a.button-like {
  background: linear-gradient(to bottom, #e2e2e2 0%, #dbdbdb 50%,
  #d1d1d1 51%, #fefefe 100%);
}
```



linear-gradient(to bottom, yellow, green);

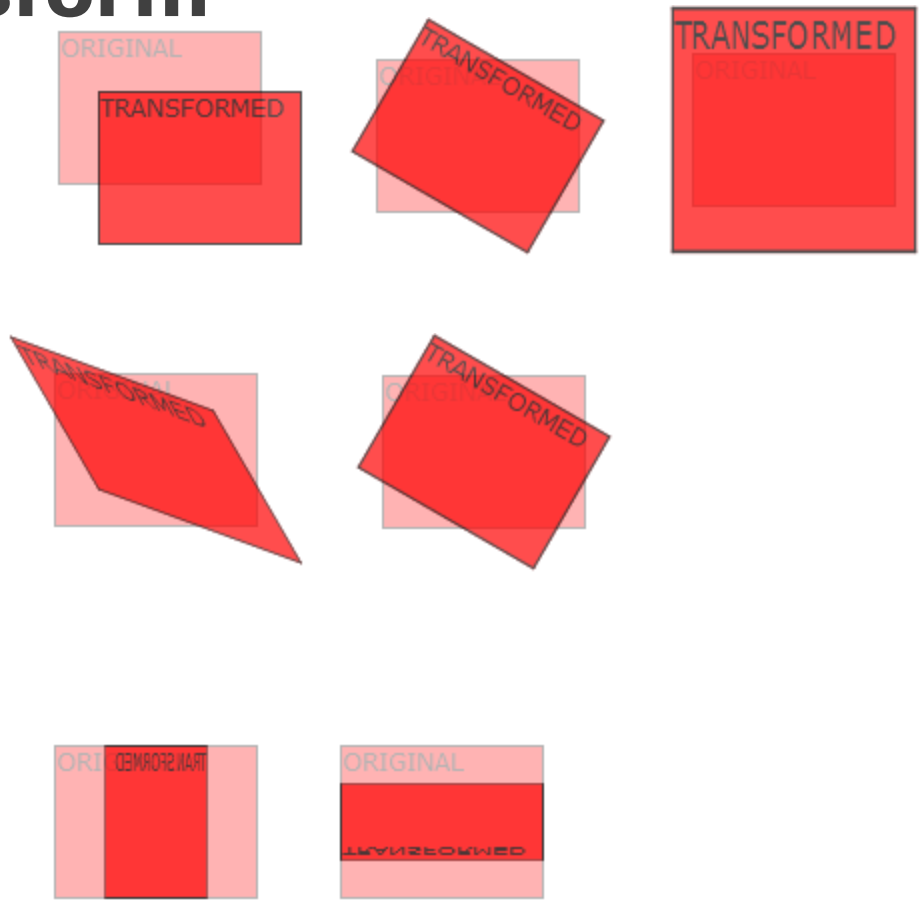


linear-gradient(90deg, yellow, orange 25%, blue);

```
background: #ffff00; /* Old browsers */
background: -moz-linear-gradient(top,  #ffff00 0%, #00ff00 100%);
    /* FF3.6+ */
background: -webkit-gradient(linear, left top, left bottom, color-
    stop(0%,#ffff00), color-stop(100%,#00ff00));
    /* Chrome,Safari4+ */
background: -webkit-linear-gradient(top,  #ffff00 0%,#00ff00 100%);
    /* Chrome10+,Safari5.1+ */
background: -o-linear-gradient(top,  #ffff00 0%,#00ff00 100%);
    /* Opera 11.10+ */
background: -ms-linear-gradient(top,  #ffff00 0%,#00ff00 100%);
    /* IE10+ */
background: linear-gradient(to bottom,  #ffff00 0%,#00ff00 100%);
    /* W3C */
filter: progid:DXImageTransform.Microsoft.gradient(
    startColorstr='#ffff00', endColorstr='#00ff00',GradientType=0 );
    /* IE6-9 */
```

Advanced CSS: Transform

- 2D Effect
  - translate()
  - rotate()
  - scale()
  - skew()
  - matrix()
- 3D Effect
  - rotateX()
  - rotateY()



Example for Transform

```
div {
   background:#F9D;
   width:350px;
   height:200px;
   transform: rotate(45deg)
}
```

```
#Window {
transform: rotate(45deg);
-ms-transform: rotate(45deg);
-moz-transform: rotate(45deg);
-webkit-transform: rotate(45de
-o-transform: rotate(45deg);
}
```

Any question?

Exercise 0

ONLINE MEDIA
HOME
STRATEGY
SOLUSI
MARKETING
WORKSHOP
OUR WORK
PRODUK
PARTNER
INSPIRASI
FAQ
Analytic
Development
Media

Bootstrap

Why We Need Front-end Framework

- Build a layout from scratch with HTML & CSS takes a lot of work and code
- We want to develop front-end faster, so the back-end coder can code the task earlier
- We build the web app for many platforms

Advantages?

- Makes life easier for developers ☺. Front-end framework greatly saves development time.
- Front-end framework usually has cutting edge interface.
- Mobile Friendly, supports all device resolution (responsive).
- Compatible with all modern browsers.

Disadvantages?

- Lots of codes.
- Many websites use it. So it could happen that your website appearance is similar to others.
- Some frameworks takes more development time because they require another environment to build the layouts and to run it

Front-end Frameworks Today

- Bootstrap
- Foundation
- TailwindCSS
- Angular
- React / Next
- Vue / Nuxt
- Svelte
- JokoUI 😘 😘 🥳 🥳

Bootstrap is..

- A framework which contains CSS library that are made specifically for web front-end development.
- Bootstrap is the world's most popular front-end framework among developers.
- Almost all web developers have used bootstrap to make the front-end development easier and faster.

Brief history of Bootstrap

Aug 2011: 1st released to public

Mark Otto
@mdo

Follow

Woohoo! Twitter Bootstrap, an open-sourced CSS/HTML framework from myself and @fat, just went live: dev.twitter.com/blog/bootstrap… #kaboom

2:25 PM - 19 Aug 2011

171 148

Search

stars:>1

Search

Repositories 1,210,476

Code

Issues

Users

Languages

JavaScript	239,019
Python	126,665
Java	108,793
Ruby	103,577
PHP	93,284
C	57,269
C++	50,711
Objective-C	42,601
Shell	34,738
C#	33,488

We've found 1,210,476 repository results

Sort: **Most stars**

twbs/bootstrap — CSS ★ 88,841 ⑂ 37,243

The most popular HTML, CSS, and JavaScript framework for developing responsive, mobile first projects on the web.

Updated an hour ago

vhf/free-programming-books — ★ 45,255 ⑂ 10,400

Freely available programming books

Updated 8 hours ago

angular/angular.js — JavaScript ★ 44,131 ⑂ 20,077

HTML enhanced for web apps

Updated 3 hours ago

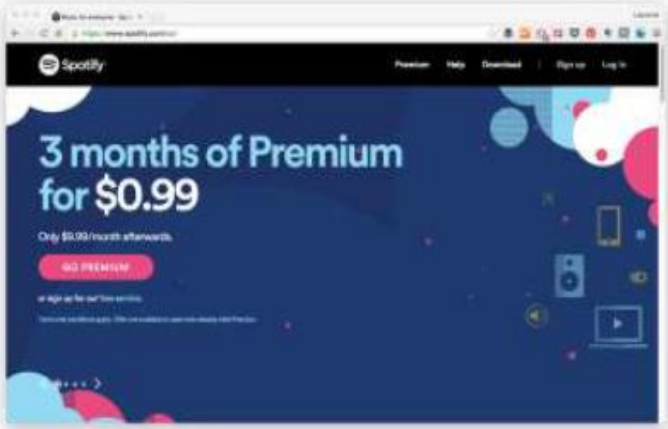
Bootstrap advantages compared to other front-end frameworks?

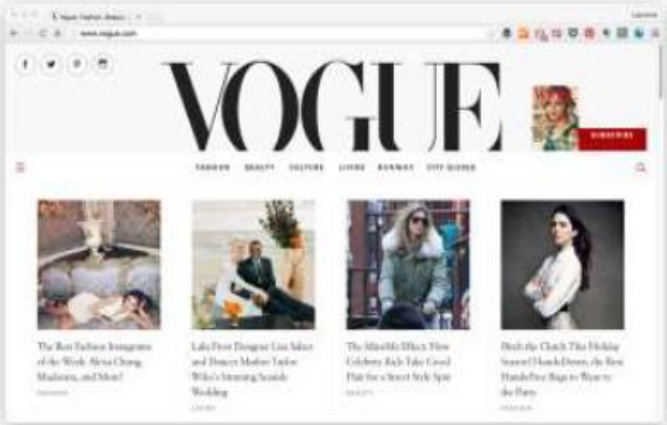
- Very light.
- Easy to learn.
- Lot of themes.
- Doesn’t require another environment to build the layouts

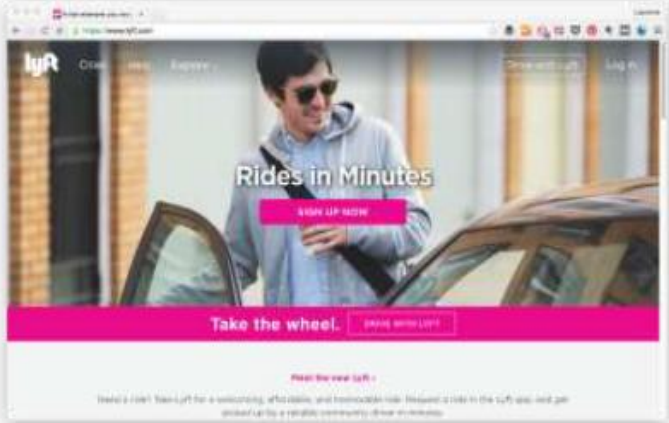
Bootstrap Disadvantages compared to other front-end frameworks?

- Extensive classes. Specific class for specific interface.
- Still using ‘old semantic rules’. But it works, so..

Sites Built With Bootstrap







When to use Bootstrap?

- Prototyping process (interface design, mock-up, etc)
- Want to develop front-end faster

Prerequisites

- To be able to use Bootstrap, you must understand:
  - HTML,
  - CSS

How to use Bootstrap

- **Uses external URL for linking the CSS and JS file, so you must connect to the internet in order to use Bootstrap**

1. Create an HTML file

PC > WINDOWS (C:) > xampp > htdocs > contoh_project

Name	Date modified	Type	Size
css	18/01/2018 8:12	File folder	
js	18/01/2018 8:12	File folder	
index.html	15/02/2018 10:39	Chrome HTML Do...	0 KB

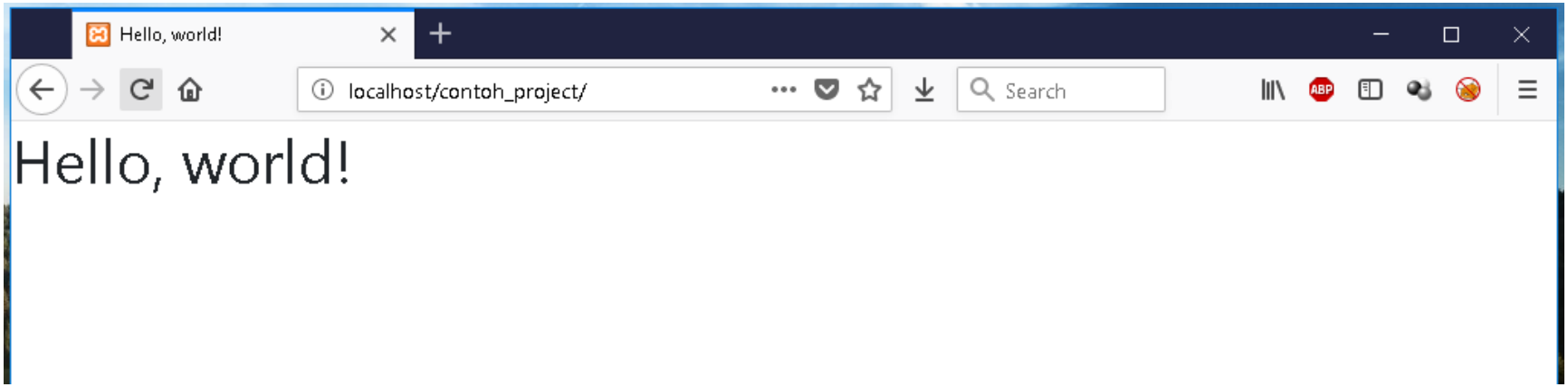
2. Try the source code template (copy and paste it to the HTML file (https://getbootstrap.com/docs/5.3/getting-started/introduction/)

</body>. Learn more about our CDN links.

Copy to clipboard

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Bootstrap demo</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/css/bootstrap.min.css
  </head>
  <body>
    <h1>Hello, world!</h1>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0-alpha1/dist/js/bootstrap.bundle.
  </body>
</html>
```

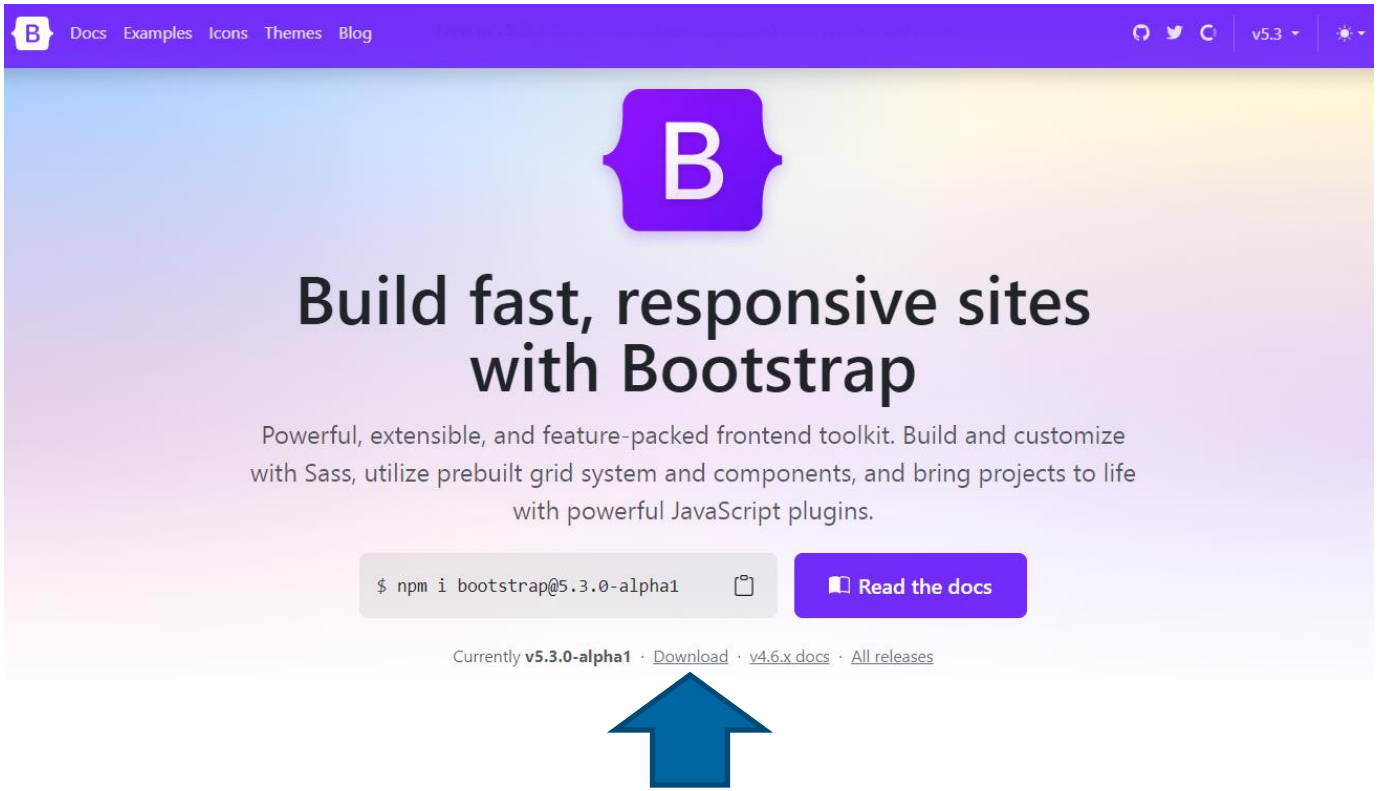
3. See the result, make sure there is no error in developer mode console. Voila, you ready to use Bootstrap



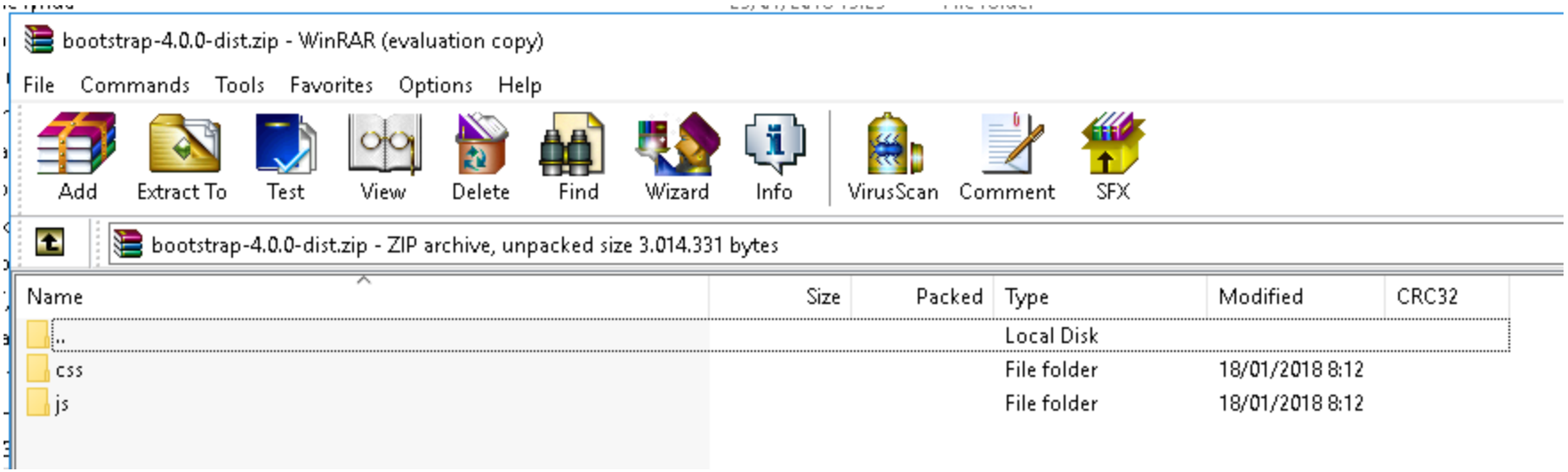
- **Use Bootstrap offline**

1. Download from

- https://getbootstrap.com/



2. Extract the file



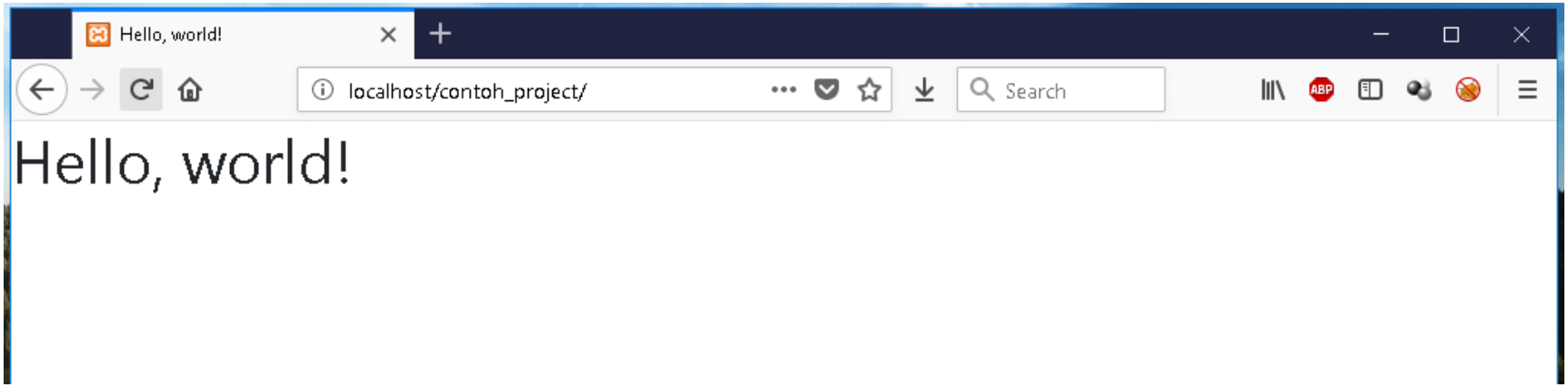
3. Open the folder, create an HTML file

PC > WINDOWS (C:) > xampp > htdocs > contoh_project

Name	Date modified	Type	Size
css	18/01/2018 8:12	File folder	
js	18/01/2018 8:12	File folder	
index.html	15/02/2018 10:39	Chrome HTML Do...	0 KB

4. Insert CSS dan JS with the path of bootstrap.css and bootstrap.js linking, by pointing it to the local Bootstrap folder you've extracted before

5. See the result, make sure there is no error in developer mode console. Voila, you ready to use Bootstrap



Using Bootstrap Components

GRID

- **Columns are incredibly flexible.** There are 12 template columns available per row, allowing you to create different combinations of elements that span any number of columns. Column classes indicate the number of template columns to span (e.g., `col-4` spans four). `widths` are set in percentages so you always have the same relative sizing.

8 + 4 = 12

.col-md-8	.col-md-4

4 + 4 + 4 = 12

.col-md-4	.col-md-4	.col-md-4

Three equal columns

Get three equal-width columns **starting at desktops and scaling to large desktops**. On mobile devices, tablets and below, the columns will automatically stack.

.col-md-4	.col-md-4	.col-md-4

Three unequal columns

Get three columns **starting at desktops and scaling to large desktops** of various widths. Remember, grid columns should add up to twelve for a single horizontal block. More than that, and columns start stacking no matter the viewport.

.col-md-3	.col-md-6	.col-md-3

Two columns

Get two columns **starting at desktops and scaling to large desktops**.

.col-md-8	.col-md-4

Full width, single column

No grid classes are necessary for full-width elements.

How to make a Grid

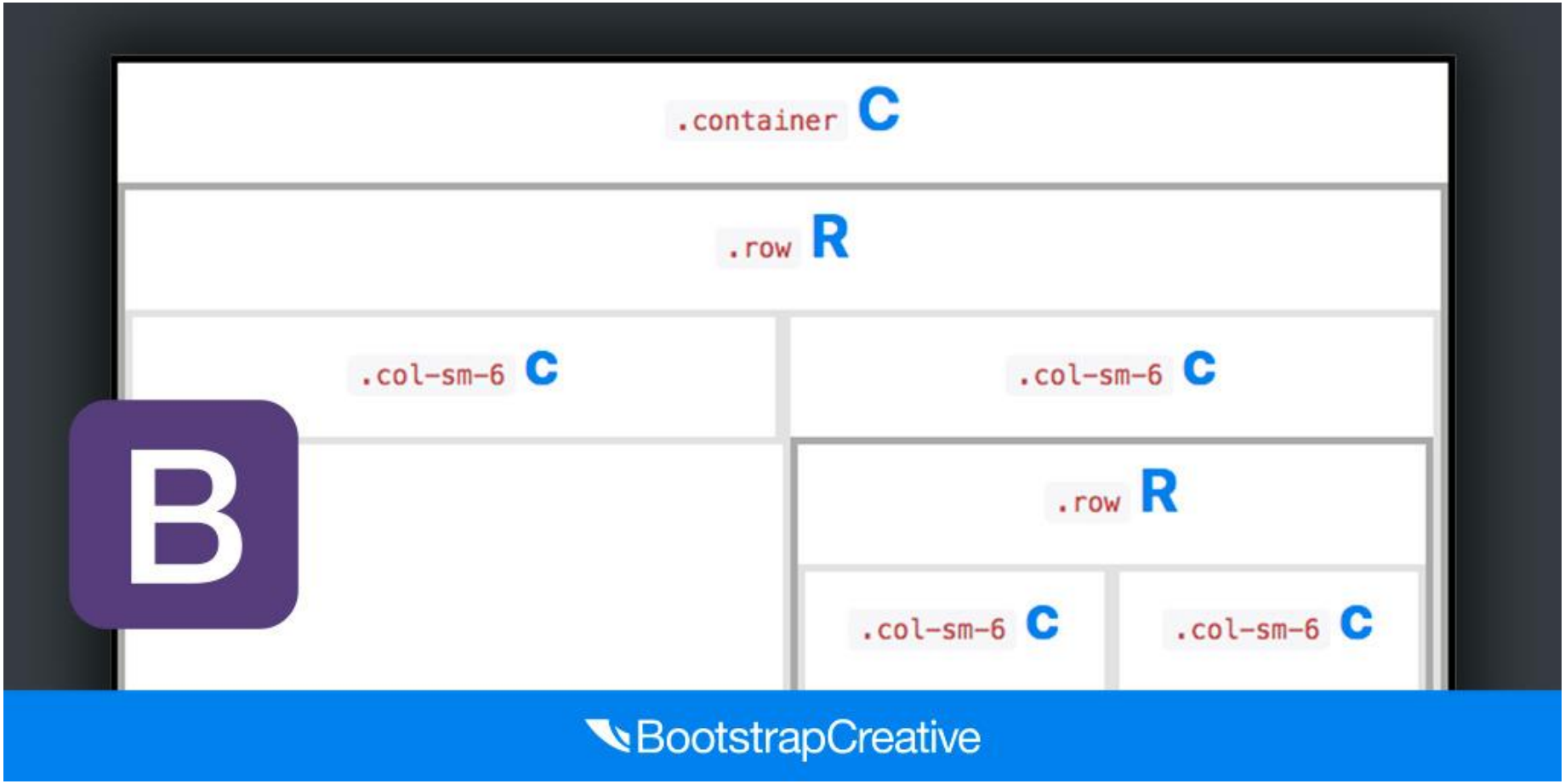
- You have to put the grid boxes (“col..” class) inside a box with “row” class:

** … **

```
<div class="row">
<div class="col-md-6">
  kolom nomer 1
</div>
<div class="col-md-6">
  kolom nomor 2
</div>
</div>
```

Grid option

	xs <576px	sm ≥576px	md ≥768px	lg ≥992px	xl ≥1200px	xxl ≥1400px
Container `max-width`	None (auto)	540px	720px	960px	1140px	1320px
Class prefix	`.col-`	`.col-sm-`	`.col-md-`	`.col-lg-`	`.col-xl-`	`.col-xxl-`
# of columns	12					
Gutter width	1.5rem (.75rem on left and right)					
Custom gutters	Yes					
Nestable	Yes					
Column ordering	Yes					


.container C
.row R
.col-sm-6 C
.col-sm-6 C
.row R
.col-sm-6 C
.col-sm-6 C
B
BootstrapCreative

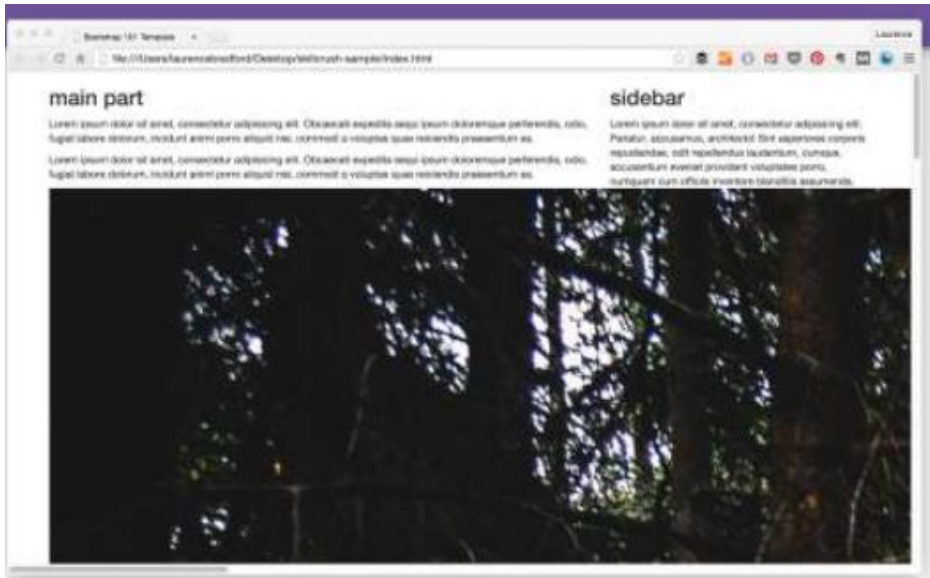
Images

```
<img src="..." class="img-thumbnail border border-dark" />
```

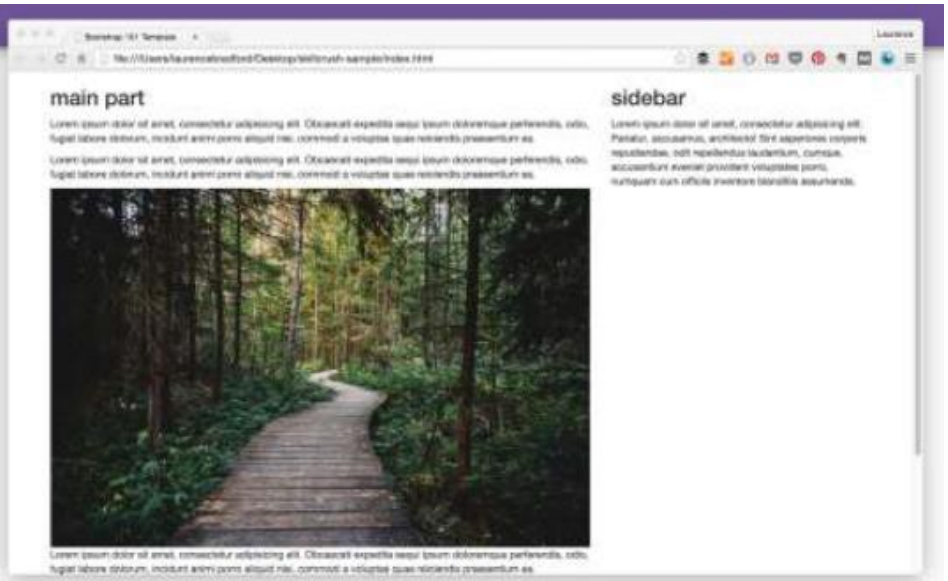
```
<img src="..." class="rounded-circle" />
```

Images (responsive)

```
<img src="..." class="img-fluid" />
```

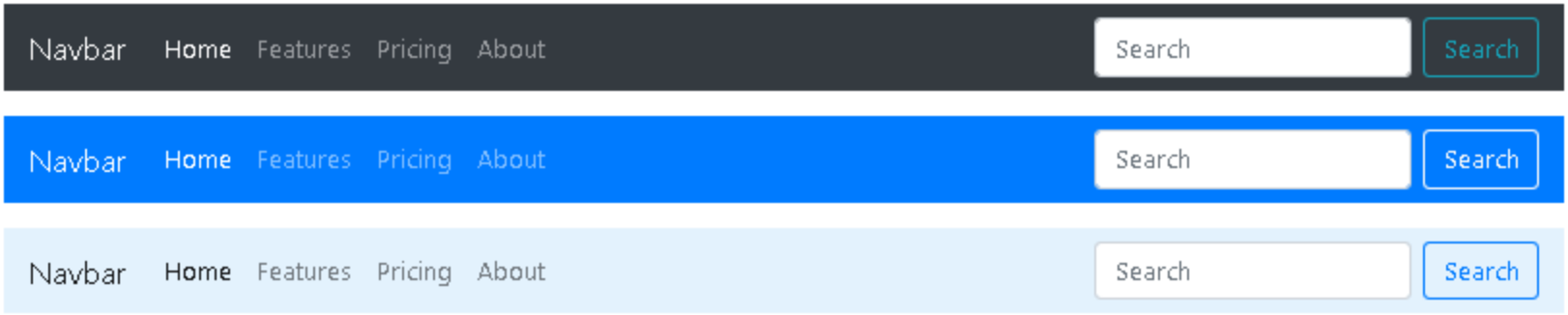


before



after

Navigation bar



Button

Primary Secondary Success Danger Warning Info Light Dark Link

Copy

```
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>

<button type="button" class="btn btn-link">Link</button>
```

Button Outline

Primary Secondary Success Danger Warning Info Light Dark

Copy

```
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-secondary">Secondary</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-light">Light</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
```

Progress Bar

Copy

```
<div class="progress">
  <div class="progress-bar" role="progressbar" aria-valuenow="0" aria-valuemin="0" aria-valuemax="100"></div>
</div>
<div class="progress">
  <div class="progress-bar" role="progressbar" style="width: 25%" aria-valuenow="25" aria-valuemin="0" aria-valuemax="100"></div>
</div>
<div class="progress">
  <div class="progress-bar" role="progressbar" style="width: 50%" aria-valuenow="50" aria-valuemin="0" aria-valuemax="100"></div>
</div>
<div class="progress">
  <div class="progress-bar" role="progressbar" style="width: 75%" aria-valuenow="75" aria-valuemin="0" aria-valuemax="100"></div>
</div>
<div class="progress">
  <div class="progress-bar" role="progressbar" style="width: 100%" aria-valuenow="100" aria-valuemin="0" aria-valuemax="100"></div>
</div>
```

25%

Copy

```
<div class="progress">
  <div class="progress-bar" role="progressbar" style="width: 25%;" aria-valuenow="25" aria-valuemin="0" aria-valuemax="100">25%</div>
</div>
```

Copy

```
<div class="progress">
  <div class="progress-bar progress-bar-striped" role="progressbar" style="width: 10%" aria-valuenow="10" aria-valuemin="0" aria-valuem
</div>
<div class="progress">
  <div class="progress-bar progress-bar-striped bg-success" role="progressbar" style="width: 25%" aria-valuenow="25" aria-valuemin="0"
</div>
<div class="progress">
  <div class="progress-bar progress-bar-striped bg-info" role="progressbar" style="width: 50%" aria-valuenow="50" aria-valuemin="0" ari
</div>
<div class="progress">
  <div class="progress-bar progress-bar-striped bg-warning" role="progressbar" style="width: 75%" aria-valuenow="75" aria-valuemin="0"
</div>
<div class="progress">
  <div class="progress-bar progress-bar-striped bg-danger" role="progressbar" style="width: 100%" aria-valuenow="100" aria-valuemin="0"
</div>
```

https://getbootstrap.com/docs/4.0/components/forms/

Input form

Email address

Enter email

We'll never share your email with anyone else.

Password

Password

☐ Check me out

Submit

Copy

```
<form>
  <div class="form-group">
    <label for="exampleInputEmail1">Email address</label>
    <input type="email" class="form-control" id="exampleInputEmail1" aria-describedby="emailHelp" placeholder="Enter email">
    <small id="emailHelp" class="form-text text-muted">We'll never share your email with anyone else.</small>
  </div>
  <div class="form-group">
    <label for="exampleInputPassword1">Password</label>
    <input type="password" class="form-control" id="exampleInputPassword1" placeholder="Password">
  </div>
  <div class="form-check">
    <input type="checkbox" class="form-check-input" id="exampleCheck1">
    <label class="form-check-label" for="exampleCheck1">Check me out</label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Pagination

Previous 1 2 3 Next

Copy

```
<nav aria-label="Page navigation example">
  <ul class="pagination">
    <li class="page-item">
    <li class="page-item">
    <li class="page-item">
    <li class="page-item">
    <li class="page-item">
  </ul>
</nav>
```

And many more components

Explore it yourself:

https://getbootstrap.com/docs/5.3/getting-started/introduction/

Examples:

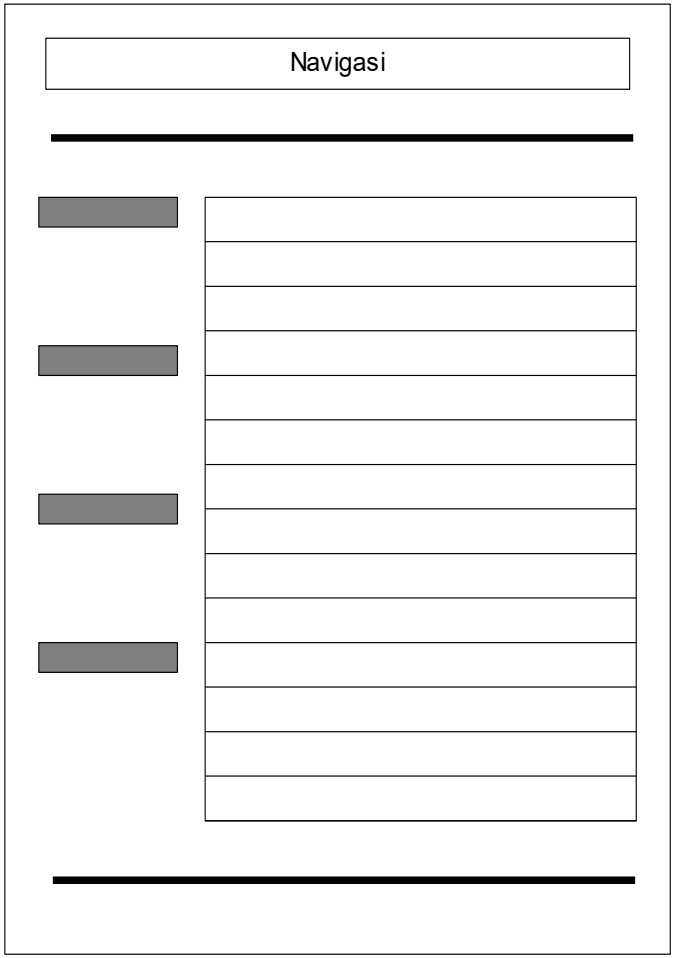
https://getbootstrap.com/docs/5.3/examples/

Any question?

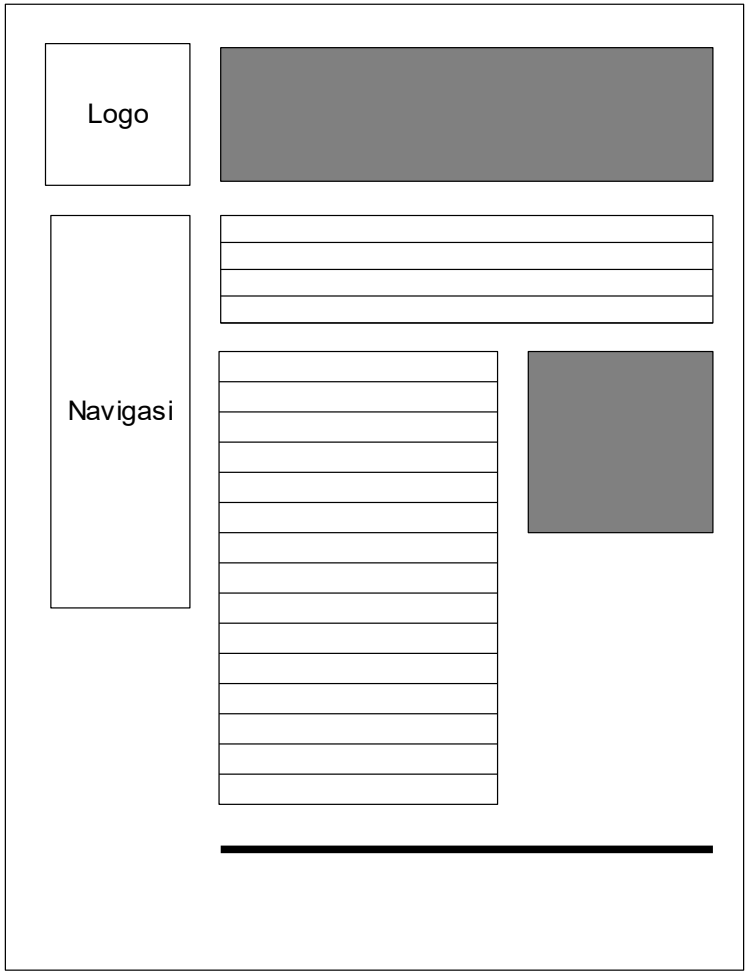
Exercise 1

ONLINE MEDIA
HOME
STRATEGY
SOLUSI
MARKETING
WORKSHOP
OUR WORK
PRODUK
PARTNER
INSPIRASI
FAQ
Analytic
Creative
Development
Marketing
Media
Optimitation

Exercise 2: Layout



Exercise 3: Advanced Layout



References

- https://www.w3schools.com/css/default.asp
- https://getbootstrap.com/docs/5.3/getting-started/introduction/
- https://www.w3schools.com/bootstrap5/default.asp

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THANK YOU